

Amirkabir University of Technology (AUT)

Department of Computer Science

Research Report

User Count Prediction Based on Advertising Expenditure and Time Series Analysis

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Chapter 1

Model Description and Assumptions

For generating synthetic data, several key assumptions were considered. The first assumption is that the daily number of application users is influenced by advertising expenditure. For this reason, the variable `Ad_Spend` is defined as the daily advertising cost. It is assumed that the higher the advertising expenditure on a given day, the greater the number of users on that same day. This effect is captured in the model by a positive coefficient for the same-day advertising cost.

The second assumption concerns the lagged effect of advertising. In the real world, a user may see an advertisement today but open the application or install it the following day. Therefore, in addition to the current day's advertising cost, the variable `Ad_Spend_Lag1` is defined to represent the previous day's advertising expenditure. This variable allows the delayed effect of advertising to be reflected in the user count. In the model, the effect of the previous day's advertising is positive, but smaller in magnitude than the same-day effect.

The third assumption is that the number of users follows a growth trend over time. Generally, as time passes, user awareness of the application increases and the user count tends to grow. To capture this, the variable `Day` is included in the model. Additionally, to prevent growth from being purely linear and unbounded, a quadratic time term, Day^2 , is also included. The coefficient of this term is negative, indicating that the rate of user growth gradually slows over time.

The fourth assumption relates to the weekend effect. In this project, it is assumed that more users engage with the application on weekends. Accordingly, the binary variable `is_weekend` is defined, taking the value 1 if the day is Thursday or Friday, and 0 otherwise. A positive effect is assigned to this

variable in the model, meaning that weekend days lead to a higher baseline user count.

The general formula for the baseline (expected mean) number of users is:

$$\text{Users_Base} = 100 + 3(\text{Ad_Spend}) + 1.5(\text{Ad_Spend_Lag1}) + 2(\text{Day}) - 0.01(\text{Day}^2) + 50(\text{is_weekend}) \quad (1.1)$$

In this formula, 100 is the baseline user level. The coefficient 3 represents the same-day advertising effect, 1.5 represents the lagged advertising effect, 2 represents the positive time trend, -0.01 represents the deceleration of growth over time, and 50 represents the positive weekend effect.

In the final step, the actual daily user count is generated using the Poisson distribution:

$$Users_t \sim \text{Poisson}(\lambda_t), \quad \lambda_t = Users_Base_t \quad (1.2)$$

The Poisson distribution is used because the user count is a count variable—it must be a non-negative integer. This distribution ensures that the final user count is not exactly equal to the baseline value, but instead fluctuates naturally around it. Therefore, the model combines the effects of advertising, time, weekends, and the inherent randomness of count data.

Chapter 2

Data Generation Method

To simulate the daily user behavior of an application, a synthetic dataset was generated over a 180-day period. The goal was to create conditions close to reality, incorporating the effects of advertising, weekends, time trends, and random user variation.

Days were indexed from 1 to 180, with day 1 assumed to be Saturday. Thursdays and Fridays were designated as weekend days. The binary variable `is_weekend` was constructed accordingly, taking the value 1 for weekend days and 0 for all others.

Next, the daily advertising expenditure variable `Ad_Spend` was generated. At least 82 days were randomly selected as no-advertising days. To make the data more realistic, approximately 40% of weekends were protected from being assigned as no-advertising days, increasing the likelihood of advertising on weekends.

Approximately 12 three-day advertising campaigns were then created. On campaign days, the advertising cost was drawn from the range 40 to 60. For other advertising days, the cost was drawn from the range 25 to 40 on weekends and from 10 to 25 on regular weekdays.

$$Ad_Spend = \begin{cases} 0, & \text{No advertising day} \\ \text{Uniform}(40, 60), & \text{Campaign day} \\ \text{Uniform}(25, 40), & \text{Non-campaign weekend} \\ \text{Uniform}(10, 25), & \text{Regular non-campaign day} \end{cases} \quad (2.1)$$

To account for the lagged advertising effect, the variable `Ad_Spend_Lag1` was constructed as the previous day's advertising cost. The baseline user

count was then computed using the formula:

$$\text{Users_Base} = 100 + 3(\text{Ad_Spend}) + 1.5(\text{Ad_Spend_Lag1}) + 2(\text{Day}) - 0.01(\text{Day}^2) + 50(\text{is_weekend}) \quad (2.2)$$

Finally, the daily user count was generated using the Poisson distribution:

$$\text{Users}_t \sim \text{Poisson}(\lambda_t), \quad \lambda_t = \text{Users_Base}_t \quad (2.3)$$

The final dataset contains the following columns:

Column Name	Data Type	Description	Role in Analysis
Day	Integer	Day index from 1 to 180	Time variable
Ad_Spend	Float	Advertising cost on the same day	Main independent variable
Ad_Spend_Lag1	Float	Previous day's advertising cost	Lagged independent variable
is_weekend	Binary	Indicates whether the day is a weekend	Control variable
Users	Integer	Generated daily user count	Dependent / target variable

Table 2.1: Final dataset column descriptions

Chapter 3

Data Analysis and Statistical Examination

3.1 Part One: Descriptive Analysis and Correlation

In this section, the data on daily application user counts and advertising expenditure over a 180-day period are examined. Based on descriptive statistics, the mean daily user count is 266.88 and the median is 257.50. The proximity of the mean and median indicates that the distribution of user counts is largely symmetric; however, a skewness value of 0.641 suggests a slight rightward tail toward higher values. A kurtosis value of -0.156 indicates that the distribution is approximately normal, with no extreme outliers or sharp peaks observed in the data. The standard deviation of user counts is 83.65, reflecting considerable day-to-day variability.

Regarding advertising expenditure, the mean daily cost is \$18.27 and the median is \$14.60. Since the mean exceeds the median, it can be inferred that advertising costs were relatively high on certain days, pulling the mean upward. Furthermore, out of the 180 days examined, 82 days had no advertising and 98 days included advertising activity. This indicates that the dataset covers both advertising and non-advertising conditions, enabling a direct comparison of the effect of advertising on user counts.

A comparison between regular weekdays and weekends reveals a notable difference in user behavior between the two groups. The mean user count on regular weekdays is 245.95, while on weekends it is 321.30. On average, weekends recorded 75.35 more users than weekdays, corresponding to an increase of approximately 30.6% relative to regular weekday levels.

The correlation analysis results also indicate that the strongest relation-

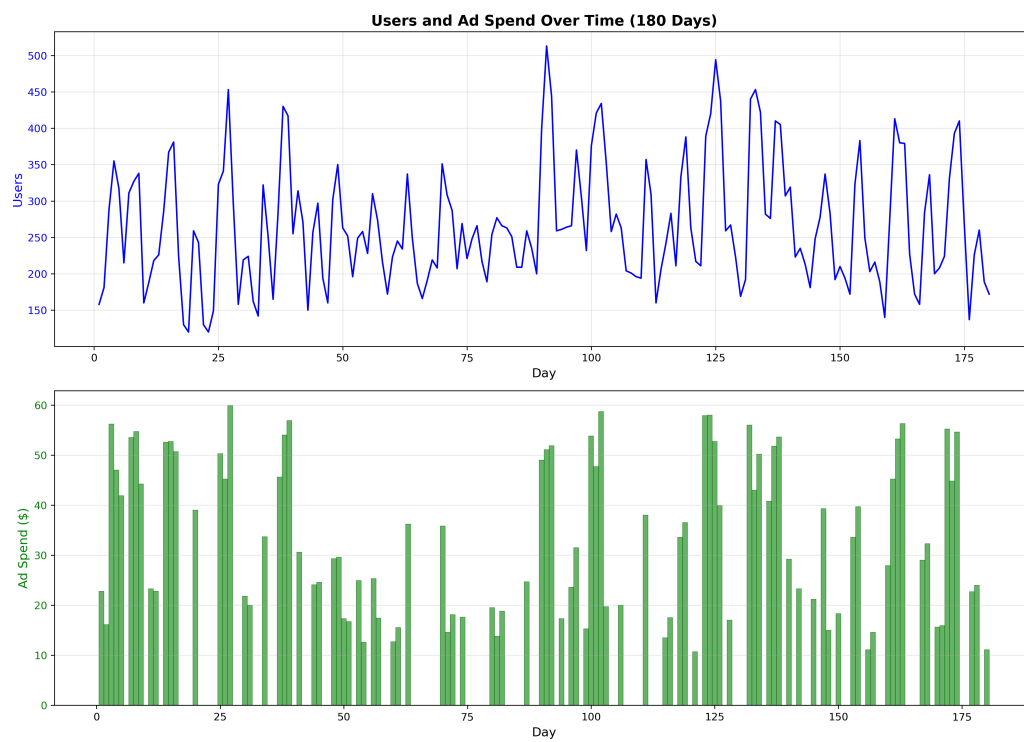


Figure 3.1: Trends in daily user count and advertising expenditure over 180 days

ship is between same-day advertising expenditure and user count. The correlation coefficient between these two variables is 0.826, representing a strong positive relationship. In other words, as advertising expenditure increases, user count increases substantially. Additionally, the previous day's advertising expenditure variable also exhibits a positive correlation of 0.553 with user count. This finding demonstrates that the effect of advertising is not limited to the same day and can also influence user count on the following day.

The weekend indicator variable also shows a positive correlation of 0.405 with user count, which is consistent with the results of the mean comparison. In contrast, the time variable, or day number, has a weak correlation of 0.108 with user count; therefore, a very mild increasing trend over time is observed in the data, but its effect is much smaller compared to advertising and weekend effects.

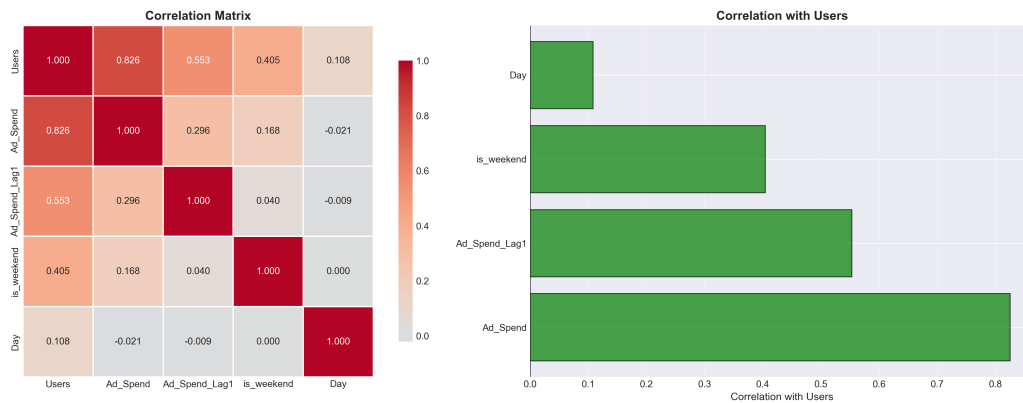


Figure 3.2: Correlation matrix

Based on this figure and the above explanations, the **Ad_Spend** variable has the strongest positive correlation with **Users**, while **Day** shows the weakest relationship; therefore, advertising variables play a more important role in predicting users.

3.2 Part Two: Analysis of Advertising Effect on User Count

In this section, the relationship between advertising expenditure and daily user count was examined. First, days were divided into four categories based on advertising level: no advertising, low advertising, medium advertising,

and high advertising. The results showed that as the advertising level increases, the average user count also increases significantly. The average user count on days without advertising was 210.22 users, while this value reached 245.10 users on days with low advertising, 334.35 users on days with medium advertising, and 383.89 users on days with high advertising. This pattern demonstrates that advertising has a direct and increasing effect on user acquisition.

For a more detailed examination, days were divided into two groups: with advertising and without advertising. The average user count on days without advertising was 210.22 users, and on days with advertising was 314.30 users. This difference represents an increase of 104.08 users and approximately 49.5% growth on days with advertising. Furthermore, the t-test result with a p-value of approximately 0 showed that this difference is statistically highly significant and cannot be attributed to chance.

Subsequently, the linear relationship between advertising and users was examined using regression. In the simple model $\text{Users} \sim \text{Ad_Spend}$, the advertising coefficient was 3.41; that is, on average, with each unit increase in advertising expenditure, approximately 3.4 users are added to the user count. The value $R^2 = 0.6815$ also indicates that about 68% of user variations are explained by same-day advertising. In the multivariate regression model that considers both today's advertising and the previous day's advertising, the coefficient for today's advertising was 3.01 and the coefficient for the previous day's advertising was 1.37, and the R^2 value increased to 0.7919. This result demonstrates that both same-day advertising and the lagged effect of the previous day's advertising play an important role in explaining user variations.

3.3 Part Three: Analysis of Weekend and Weekday Effects on User Count

In this section, the impact of weekends and different days of the week on user count was examined. In this project, Thursday and Friday were considered as the weekend. The results showed that the average user count on regular weekdays is 245.95 users and on weekends is 321.30 users. Therefore, on weekends there are on average 75.35 more users, equivalent to a 30.6% increase compared to regular days. The t-test with a p-value of approximately 0 also showed that this difference is statistically significant.

Day-by-day analysis of the week also revealed that user behavior is not uniform across all days of the week. The highest average user count be-

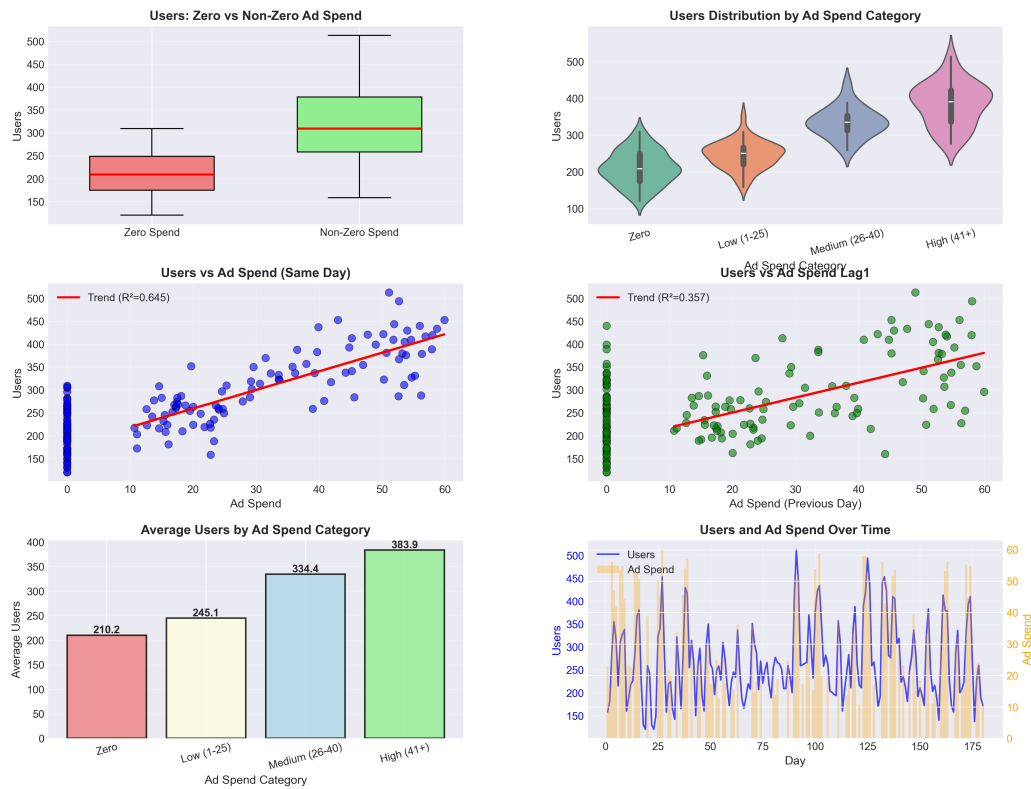


Figure 3.3: Advertising effect analysis (6 panels): users comparison by advertising presence, distribution by advertising category, scatter plots of users vs. same-day and lagged advertising with trend lines, average users by category, and time series of users and advertising expenditure

longs to Friday (330.60), followed by Thursday (312.00). In contrast, the lowest average belongs to Monday (238.88). Saturday, Sunday, Tuesday, and Wednesday are also in a close range, with their averages mainly between 241 and 255 users. This difference indicates that there is a distinct weekly pattern in user behavior, and users are more active in the application during the final days of the week.

Distribution plots also confirm this result. The distribution of users on weekends is shifted toward higher values compared to regular days, and its mean is also larger. Additionally, in time series plots, many peaks in user count coincide with weekend periods. This indicates that being a weekend is an important variable in predicting user count and should be considered as one of the main features in modeling.

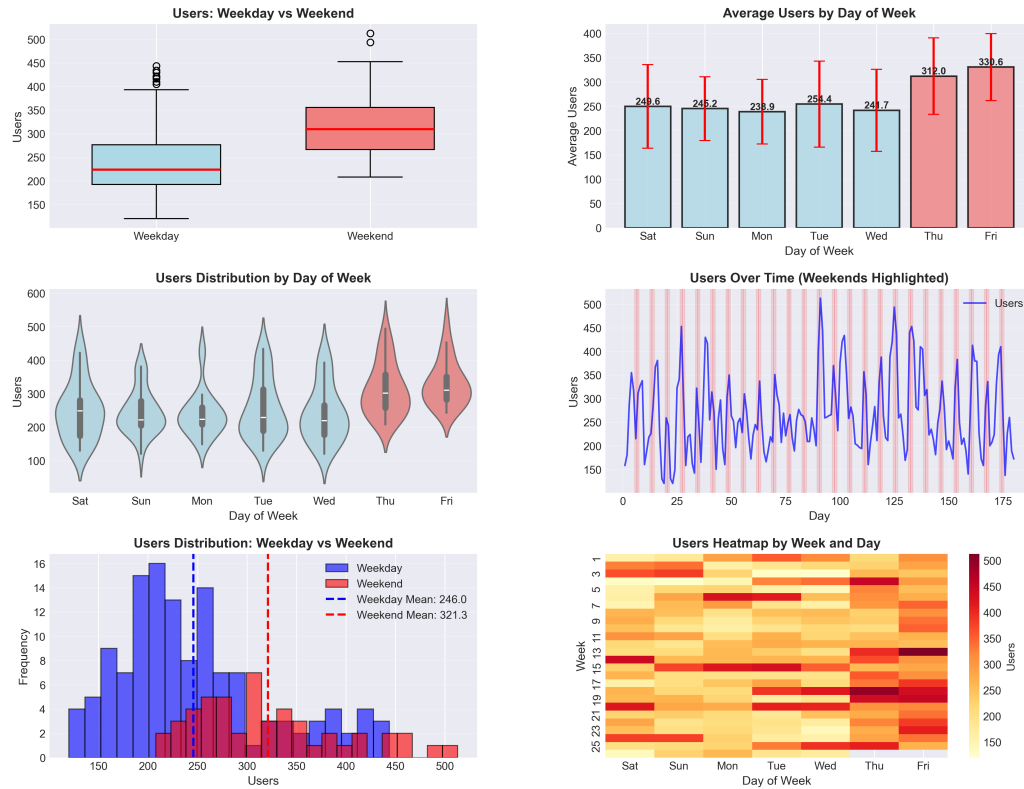


Figure 3.4: Weekend effect analysis (6 panels): weekday vs. weekend comparison, average users by day of week, distribution by day of week, time series with weekends highlighted, distribution comparison histogram, and heatmap by week and day

Chapter 4

Modeling

4.1 Part One: Time Series Analysis of User Count

In this section, the behavior of user count was examined as a time series to better identify trends, recurring patterns, and the degree of data fluctuations. The results of time series decomposition show that user count consists of three main components: trend, weekly seasonality, and residual or noise. The trend component varies between 164.43 and 384.00, indicating the presence of an overall upward trend during the 180-day period. This indicates that alongside daily fluctuations, the overall level of users has grown in some intervals. The seasonal component also ranges from -33.68 to $+61.59$, confirming the existence of a regular weekly pattern. Additionally, the standard deviation of the residual component is 63.75, which shows that part of the user variations is due to random factors or variables not explained in the model.

To examine the stationarity of the time series, the ADF test was performed on the user count variable. The test statistic value was -3.5558 and the p-value was 0.0067, which is less than 0.05; therefore, the non-stationarity hypothesis is rejected and it can be said that the user time series is stationary at an appropriate confidence level. Furthermore, after differencing, the time series remained stationary and the p-value became approximately zero. This result is very important from a modeling perspective, as many time series forecasting models depend on data stationarity, and this characteristic makes the modeling process more reliable.

Examination of the 7-day moving average shows that the overall trend of users over time, despite daily fluctuations, is observable and relatively smooth. In some intervals, especially around the middle and later days of the period, the moving average level has increased and then returned to

lower levels. On the other hand, the 7-day moving standard deviation also indicates that the intensity of fluctuations has not been constant throughout the entire period. The average of this index is 75.67, which shows that user variations in different intervals have been relatively significant but generally controllable. Therefore, it can be said that the data, in addition to trend and weekly pattern, have fluctuations of moderate intensity.

Lag analysis also provides important information about the temporal dependency of the data. The highest correlation is observed between today's user count and the previous day's user count, such that the correlation coefficient at lag 1 is 0.5109. This result shows that each day's user count is largely dependent on the previous day's condition. At lags 2 to 6 days, weak and mostly negative correlation is observed, indicating a decrease in the direct effect of more distant days. At lag 7, the correlation becomes positive again and reaches 0.1141. Although this value is not large, alongside seasonal plots and weekly day patterns, it shows that a weekly cycle exists in user behavior.

Analysis of the weekly pattern also reinforces this result. The average number of users on different days of the week is not uniform and a significant difference is observed between them. The highest average belongs to Friday with 330.60 users, followed by Thursday with 312.00 users, while the lowest value belongs to Monday with 238.88 users. The difference between the highest and lowest values is about 91.72 users, which shows that the type of weekday has a clear impact on the level of application usage. Additionally, daily distribution plots show that on the final days of the week, not only is the average number of users higher, but the overall level of distribution has also shifted to higher values.

Overall, time series analysis shows that the application's user count has a mild upward trend, a weekly seasonal pattern, and dependency on previous day values. These findings confirm the importance of using features such as lag 1, day of week, and temporal components in the user prediction model. Therefore, for more accurate prediction of user count in future days, the effects of trend, weekly pattern, and short-term dependency between consecutive days must be simultaneously incorporated into the model.

Based on the analysis results, this time series is of the univariate type with exogenous factors; because the main variable to be predicted is daily user count, but other variables such as advertising cost, previous day's advertising cost, being a weekend, and day of week also affect it. On the other hand, the data has a clear weekly pattern, dependency on past values, and a mild upward trend. Therefore, this time series is neither completely random nor only time-dependent, but rather contains a combination of temporal structure and external influencing variables.

Given these characteristics, the most appropriate choice for modeling is

time series models with exogenous variables. Among them, the SARIMAX model is considered one of the best options for this project. The reason for this choice is that SARIMAX can simultaneously model autocorrelation between days, weekly seasonal pattern, and the effect of external variables such as advertising and weekends.



Figure 4.1: Time series decomposition: original series, trend component, seasonal component (weekly), and residual component over 180 days

As evident from the plot, the user time series has significant fluctuations, but the overall trend becomes upward in the middle of the interval and then returns to a more stable range.

The weekly seasonal pattern is completely clear; peaks and troughs are repeated regularly and indicate the periodic behavior of users on weekends.

4.2 Part Two: Comparison of Regression Models and SARIMAX for User Prediction

In this section, the performance of three different models for predicting user count was examined and compared: simple Linear Regression, Linear Regression with the Day^2 variable, and the SARIMAX model. The goal of this comparison is to select a model that can predict actual user values with the least error and highest accuracy.

4.2.1 First Model: Linear Regression

Based on the obtained results, the Linear Regression model has an R^2 value of 0.9009. This value indicates that the model can explain approximately 90% of the variations in user count. Although this number is acceptable, the model's error value is relatively high. The RMSE value of 25.60 and MAE of 19.91 show that this model's predictions have, on average, greater differences from actual values. In the scatter plot, it is also observed that the points have greater distance from the red reference line.

4.2.2 Second Model: Linear Regression + Day^2

In the second model, Linear Regression + Day^2 , in addition to the main variables, the nonlinear effect of time has also been considered by adding the Day^2 variable. This change has led to significant improvement in model performance. The R^2 value has increased to 0.9619 and the RMSE value has decreased to 15.86. Also, the MAE value is 12.51, which shows a noticeable decrease compared to the simple linear model. This result indicates that the relationship between time and user count is not completely linear, and adding the second-degree term has been able to better model the data trend.

4.2.3 Third Model: SARIMAX

The third model, SARIMAX, is one of the time series models that, in addition to temporal dependency of data, also provides the possibility of using exogenous variables. This model has performed very close to the second model in this problem. The R^2 value of 0.9605, RMSE of 15.86, and MAE of 12.52 have been reported. These numbers show that SARIMAX also has very high accuracy in predicting user count and is particularly a suitable choice for data with temporal patterns. In the SARIMAX(2,1,2)(1,0,1,7) model, the first part, (2,1,2), respectively indicates the number of autoregressive

lags, the degree of differencing for stationarity, and the number of moving averages, and the seasonal part, (1,0,1,7), also states that the model has one seasonal moving average component and a 7-day seasonal period.

4.2.4 Examination of the Best Model's Coefficients

According to the results, the Linear Regression + Day² model has had the best performance among the three models, as it has provided the highest R^2 value and the lowest MAE value. The coefficients of this model show that:

Feature	Original Coefficient (in formula)	Estimated Coefficient	Error Percentage
Intercept (constant value)	100	94.6987	-5.30%
Ad_Spend	3	2.9793	-0.69%
Ad_Spend_Lag1	1.5	1.5490	3.26%
Day	2	2.1204	6.02%
is_weekend	50	50.1104	0.22%
Day ²	-0.01	-0.0106	-6.12%

Table 4.1: Coefficients of the Linear Regression + Day² model

For evaluating the accuracy and efficiency of the models used in predicting user count, three different models including simple linear regression, linear regression with the second-degree time term (Day²), and the SARIMAX model were compared with each other. This comparison was performed based on three main metrics: MAE, RMSE, and R^2 .

Model	MAE	RMSE	R^2
Linear Regression	19.91	25.60	0.9009
Linear Regression + Day ²	12.51	15.86	0.9619
SARIMAX	12.52	15.86	0.9605

Table 4.2: Comparison of model performance metrics

Overall, the results show that although the simple linear regression model has acceptable performance, its accuracy is lower compared to the other two models. Adding the Day² variable has led to significant improvement in the

model and has shown that the user growth pattern has a nonlinear trend. On the other hand, SARIMAX also has performance very close to the second model and is a very suitable option for analyzing and predicting temporal data.

Finally, according to numerical criteria, the Linear Regression + Day² model is selected as the best model, as with MAE = 12.505, RMSE = 15.865, and $R^2 = 0.9619$, it has provided the best balance between accuracy and error.

4.2.5 Examination of Charts

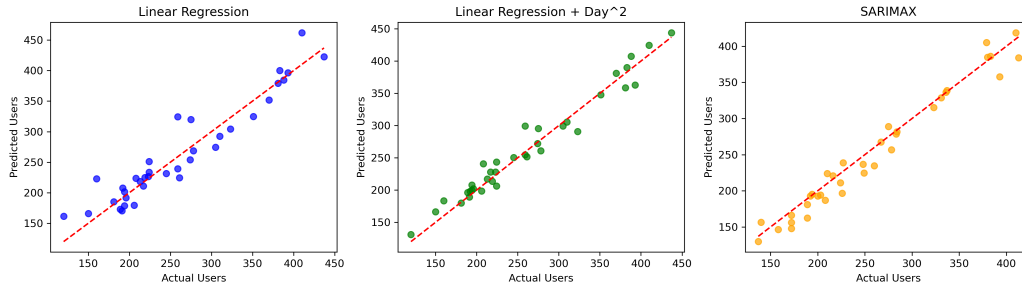


Figure 4.2: Comparison of actual vs. predicted users for three models: Linear Regression, Linear Regression + Day², and SARIMAX. The closer the points are to the red line, the higher the model accuracy.

These charts show the comparison of actual and predicted user values for the three models; the closer the points are to the red line, the higher the model accuracy.

The Linear Regression + Day² and SARIMAX models have better fit with actual data compared to simple linear regression and show less prediction error.

These charts show the residuals of the three models, and the more randomly and closer to the zero line the points are scattered, the better the model performance.

Compared to simple linear regression, the Linear Regression + Day² and SARIMAX models have more balanced scatter and more controlled errors, indicating their better fit.

This chart shows the performance of the SARIMAX model in predicting user count in the test section.

The blue line shows training data, the green line shows actual test values, and the orange line shows model predictions; the red line is the separation point between training and test data.

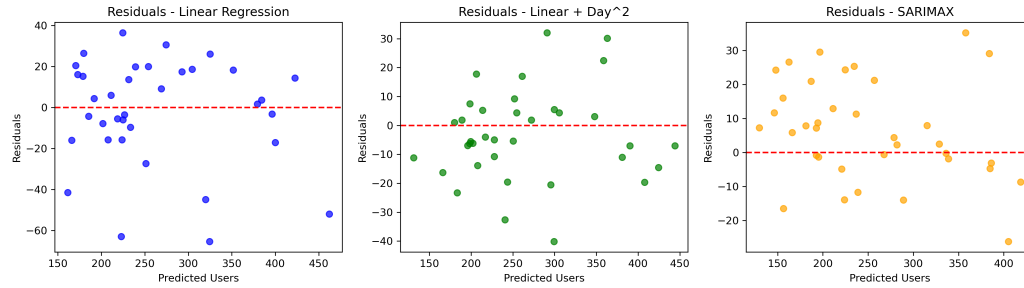


Figure 4.3: Residual plots for three models. The more randomly and closer to the zero line the points are scattered, the better the model performance.

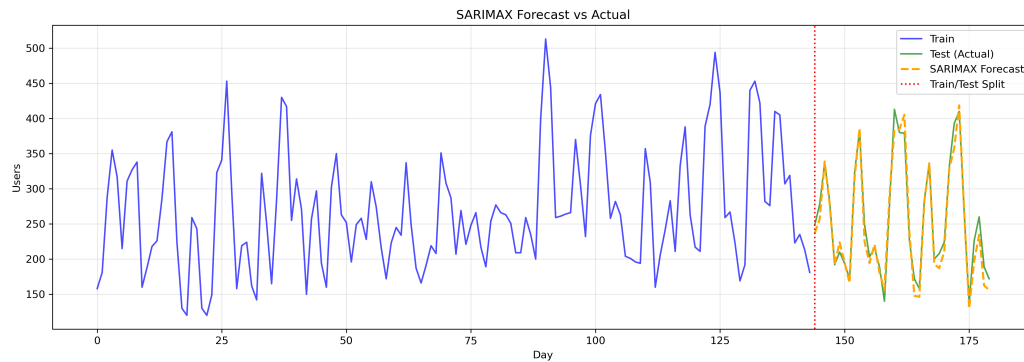


Figure 4.4: SARIMAX model performance in predicting user count in the test section. Blue line: training data, green line: actual test values, orange line: model predictions, red line: train/test split point.

Final Conclusion

The main objective of this project was to predict user count based on temporal data and influential variables such as advertising expenditure, the effect of previous day's advertising, and weekend status. In this regard, efforts were made to extract an appropriate pattern for estimating user behavior using different models, so that it could be used for analyzing trends and making more accurate decisions.

The data used in this project included variables, each of which could explain part of user behavior. The `Day` variable represented the temporal dimension of the data and helped the model identify the trend of user changes over time. The `Ad_Spend` variable indicated advertising expenditure on the same day, and the `Ad_Spend_Lag1` variable showed the residual effect of the previous day's advertising. Also, the `is_weekend` variable was considered to examine the difference in user behavior on weekends. The results showed that all these variables played an effective role in improving prediction quality, and removing any of them could deprive the model of important information. Finally, the user count was generated using the Poisson distribution.

One of the important points of this project was attention to the time series nature of the data. Since user count changes over days, merely using a simple linear relationship cannot adequately explain all existing patterns. In time series data, trend, gradual changes, lag effects, and time-dependent patterns are usually very important. In this project, it was also observed that time is not just a simple variable, but user behavior over time has a kind of nonlinear change. For this reason, adding the `Day2` term to the model was able to better capture this nonlinear behavior and significantly improve model performance. This shows that in analyzing time-dependent data, attention to the temporal structure of data and hidden patterns in it plays a very important role in the final accuracy of the model.

For modeling, three main approaches were examined: simple Linear Regression, Linear Regression with the second-degree time term, and the SARI-MAX model. Simple Linear Regression was chosen as the baseline model to determine to what extent the variables are capable of explaining user vari-

ations in the simplest case. The results of this model were acceptable, but the error value remained relatively high, and it became clear that this model cannot adequately cover all the complexities of the data.

In the next step, by adding the Day^2 variable, the model was able to better understand the temporal trend, and this led to a significant reduction in error and an increase in R^2 value. This result showed that the relationship between time and user count is not completely linear, and part of the data behavior requires a more flexible model. Then the SARIMAX model was used, which, in addition to considering influential variables, also takes into account the temporal structure of data and the dependency between consecutive observations. This model also had very good performance, and its MAE and RMSE values were at a very appropriate level, although in terms of R^2 value it was slightly lower than the Linear Regression + Day^2 model.

Examination of the final model's coefficients showed that advertising expenditure has a direct and positive effect on increasing user count; meaning that with an increase in advertising budget, user count also increases. Also, the `Ad_Spend_Lag1` variable confirmed that the effect of advertising is not limited to the same day, and part of it remains on the next day as well. The `is_weekend` variable also had a positive effect on user count, which shows that user behavior on weekends is different from regular days, and there is usually more presence or activity. On the other hand, the presence of the Day^2 term in the model clearly showed that user growth over time does not continue uniformly and linearly, and its changes have a nonlinear pattern.

Overall, it can be concluded that the combination of temporal and advertising features plays a very important role in accurately predicting user count. This project showed that using more advanced models compared to simple Linear Regression, especially in data with temporal nature, can significantly increase prediction accuracy.